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United States Patent	12387704
Kind Code	B2
Date of Patent	August 12, 2025
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System, apparatus, and method for absorbing sound

Abstract

A method is disclosed. The method includes providing a lighting assembly including at least one lighting element, attaching at least one attachment assembly to the lighting assembly, forming an empty space between the lighting assembly and a sound absorbing assembly based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member, and illuminating a translucent or transparent layer using the at least one lighting element.

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Appl. No.:	18/971814
Filed:	December 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250201224 A1	Jun. 19, 2025

Related U.S. Application Data

us-provisional-application US 63610893 20231215

Publication Classification

Int. Cl.: G10K11/16 (20060101); F21V3/06 (20180101); F21V17/12 (20060101); F21V19/00 (20060101); F21V23/06 (20060101); F21V29/70 (20150101); F21V29/89 (20150101);

U.S. Cl.:

CPC **G10K11/16** (20130101); **F21V3/06** (20180201); **F21V17/12** (20130101); **F21V19/0055** (20130101); **F21V23/06** (20130101); **F21V29/70** (20150115); **F21V29/89** (20150115); F21Y2115/10 (20160801)

Field of Classification Search

CPC: G10K (11/16); F21V (3/06); F21V (17/12); F21V (19/0055); F21V (23/06); F21V (29/70); F21V (29/89)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional patent application No. 63/610,893 filed Dec. 15, 2023, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure generally relates to a system, apparatus, and method for controlling

sound, and more particularly to a system, apparatus, and method for absorbing in conjunction with the provision of lighting.

BACKGROUND

(2) Backlight lighting fixtures typically include translucent material that is illuminated by lighting fixtures disposed behind the translucent material. The lighting fixtures thereby provide a backlight for the translucent material. In certain applications, translucent material is also transmissive to sound waves.

(3) Conventional backlight lighting fixtures include a backlighting space formed between the lighting fixtures and the translucent material that is illuminated by the lighting fixtures. This configuration usually leads to problems caused by echoing sound. Sound that enters the backlighting space from an occupied space such as an occupied room typically echoes back from the backlighting space to the occupied space.

(4) Conventional techniques exist for absorbing sound in the backlighting space to attempt to prevent echoing of sound back from the backlighting space to an occupied space. However, such conventional techniques typically limit the effectiveness of the sound absorption, create heat transfer challenges for the lighting fixture, and are difficult to install. Accordingly, a need in the art exists for efficiently absorbing sound in lighting fixtures.

(5) The exemplary disclosed system, apparatus, and method of the present disclosure are directed to overcoming one or more of the shortcomings set forth above and/or other deficiencies in existing technology.

SUMMARY OF THE DISCLOSURE

(6) In one exemplary aspect, the present disclosure is directed to a method. The method includes providing a lighting assembly including at least one lighting element, attaching at least one attachment assembly to the lighting assembly, forming an empty space between the lighting assembly and a sound absorbing assembly based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member, and illuminating a translucent or transparent layer using the at least one lighting element.

(7) In another aspect, the present disclosure is directed to a lighting apparatus for illuminating a translucent or transparent layer. The lighting apparatus includes a lighting assembly including at least one lighting element configured to illuminate the translucent or transparent layer, a sound absorbing assembly including at least one sound absorbing member, and at least one attachment assembly attached to the lighting assembly and the sound absorbing assembly so that an empty space is formed between the lighting assembly and the sound absorbing assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a 2-dimensional side view of an exemplary system of the present invention in a surface mount configuration;

(2) FIG. 2 is a 2-dimensional bottom view of an exemplary system of the present invention;

(3) FIG. 3 is a 2-dimensional detailed side view of an exemplary apparatus of the present invention in a surface mount configuration;

(4) FIG. 4 is a 2-dimensional detailed side view of a mounting point portion of the exemplary apparatus illustrated in FIG. 3;

(5) FIG. 5 is a 2-dimensional side view of another exemplary embodiment of an exemplary apparatus of the present invention;

(6) FIG. 6 is a 2-dimensional bottom view of an exemplary apparatus of the present invention;

(7) FIG. 7 is a 2-dimensional side view of another exemplary embodiment of an exemplary

apparatus of the present invention for a pendant mount configuration;

(8) FIG. 7A is a schematic view of an exemplary embodiment of an exemplary system of the present invention;

(9) FIG. 8 is a 2-dimensional side view of another exemplary embodiment of an exemplary apparatus of the present invention; and

(10) FIG. 9 illustrates an exemplary process of installing and using at least some exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION AND INDUSTRIAL APPLICABILITY

(11) The exemplary disclosed system, apparatus, and method may include one or more lighting apparatuses that may illuminate a translucent layer. For example, the one or more lighting apparatuses may be disposed in a backlighting space that is separated from an occupied space by the translucent layer, for example as described below. The exemplary disclosed system, apparatus, and method may be used in any suitable lighting configurations, for example as described below.

(12) In at least some exemplary embodiments and as illustrated in FIGS. 1 and 2, a lighting system **100** may include one or more lighting apparatuses **105**. Lighting apparatuses **105** may be disposed in a backlighting space **110** that may be separated from an occupied space **115** via a translucent layer **120**.

(13) Translucent layer **120** may be a translucent and/or transparent layer through which light can pass and that may be thereby illuminated by lighting apparatus **105**. Translucent layer **120** may be a translucent, transparent, and/or sound penetrable layer such as a translucent and sound penetrable film. Translucent layer **120** may be formed from any suitable translucent material such as, for example, vinyl material, polyester material, translucent paper material, polycarbonate material, material (e.g., polyethylene terephthalate), and/or any other suitable material for allowing light to pass through translucent layer **120**. Translucent layer **120** may have any suitable thickness such as, for example, up to about 5 mm (e.g., up to about 3 mm). In at least some exemplary embodiments, translucent layer **120** may be a translucent ceiling film. Translucent layer **120** may also be a translucent wall film, a translucent film for a lighted sign, or any other desired surface that may be illuminated. Translucent layer **120** may include micro-perforations to facilitate the passage of sound through translucent layer **120**. For example, translucent layer **120** may be a micro-perforated ceiling or wall film (e.g., or a film for any other desired application or orientation to be illuminated).

(14) Translucent layer **120** may separate backlighting space **110** from occupied space **115**. For example, occupied space **115** may be a room or other area occupied by users that may be illuminated by lighting apparatus **105** via illumination through translucent layer **120**. In at least some exemplary embodiments, occupied space **115** may be a room having a ceiling (e.g., or wall or floor) formed partially (e.g., along with glass or other transparent structural material) or substantially entirely by translucent layer **120**. For example as illustrated in FIG. 1, backlighting space **110** may be formed between translucent layer **120** and one or more support structures (e.g., support surfaces) that may support lighting apparatuses **105** (e.g., and translucent layer **120**). One or more lighting apparatuses **105** may be disposed within backlighting space **110**. Also in at least some exemplary embodiments, lighting system **100** may be used in a lighting fixture that may be independent from a ceiling (e.g., lighting system **100** may be surface-mounted below a hard ceiling or as a pendant mounted below a hard ceiling). For example, lighting apparatus **105** may be enclosed on a frame with translucent film at the bottom, and the whole assembly may then be mounted to a ceiling or suspended below a ceiling.

(15) As illustrated in FIGS. 1 and 2, lighting apparatuses **105** may be connected via one or more electrical connectors **125**. Electrical connectors **125** may be any suitable connector for transferring electricity, electrical signals, and/or data to and from lighting apparatuses **105** such as, for example, electrical lines. Electrical connectors **125** may connect lighting apparatuses **105** to a power source **130** and/or a driver **135**. In at least some exemplary embodiments, driver **135** may be a remote

driver. The exemplary disclosed power source may be any suitable LED driver. Also for example, power source **130** may be any suitable source for electrically powering lighting apparatuses **105** such as, for example, an electrical outlet, an electrical generator, a motor assembly, and/or any other suitable source of power. In at least some exemplary embodiments, lighting apparatuses **105** may be powered via wireless transmission of energy. Driver **135** may be any suitable component (e.g., an electronic driver) for regulating a variable operation and providing power (e.g., from power source **130**) to one or more lighting apparatuses **105**. For example, driver **135** may rectify voltage (e.g., relatively higher, alternating current voltage provided from power source **130**) to relatively lower (e.g., direct voltage) for powering lighting apparatuses **105**. One or more (e.g., a plurality of) lighting apparatuses **105** may be driven by driver **135**.

(16) FIG. 3 illustrates an exemplary embodiment of lighting apparatus **105**, in which a lighting element may be isolated from a sound-absorbing material, with lighting apparatus **105** being attached (e.g., mounted, for example directly mounted) to a support structure disposed above lighting apparatus **105**. Lighting apparatus **105** may be attached to a support structure **140**. Support structure **140** may include any suitable structural member, structural members, and/or structural assembly or assemblies for supporting lighting apparatus **105**. For example, support structure **140** may include one or more structural members and/or a structural assembly of a ceiling, a wall, and/or any other suitable portion of a building, transportation structure (e.g., a bridge), vehicle (e.g., ground, air, or maritime vehicle), or any other desired object or structure.

(17) Lighting apparatus **105** may include a lighting assembly **200**, a sound absorbing assembly **300**, and one or more attachment assemblies **400**. One or more attachment assemblies **400** may attach lighting assembly **200**, sound absorbing assembly **300**, and support structure **140**. For example, one, two, three, four or more (e.g., up to 10 or more) attachment assemblies **400** may attach lighting assembly **200**, sound absorbing assembly **300**, and support structure **140**. Attachment assembly **400** may attach lighting assembly **200** and sound absorbing assembly **300** so that an empty space **145** may be formed between lighting assembly **200** and sound absorbing assembly **300**. Attachment assembly **400** may also attach sound absorbing assembly **300** to support structure **140** so that a second empty space **150** may be formed between sound absorbing assembly **300** and support structure **140**. A space may alternatively be formed between the exemplary disclosed lighting assembly and the exemplary disclosed support structure for example as described in other exemplary embodiments below. Empty space **145** may provide for the exemplary disclosed attachment of lighting apparatus **105** without using a “surround” or similar structure to hold sound absorbent material. Empty space **145** may thereby provide increased exposed sound absorption surface area to increase noise reduction.

(18) Lighting assembly **200** may include a controller **205**, one or more lighting elements **210**, and a mounting member **215**. Mounting member **215** may support controller **205** and one or more lighting elements **210**.

(19) Controller **205** may control an operation of lighting apparatus **105** (e.g., including lighting assembly **200**). Controller **205** may include for example a processor (e.g., micro-processing logic control device) or board components. Also for example, controller **205** may include input/output arrangements that allow it to be connected (e.g., via wire such as electrical connectors **125**, wireless, Wi-Fi, Bluetooth, or any other suitable communication technique) to other devices and components of lighting system **100**. Controller **205** may communicate with any suitable user interface such as a control panel including buttons and switches, a keypad, a touchscreen, voice control interface, and/or any other suitable user interface. For example, controller **205** may communicate with driver **135**. Controller **205** may control an operation of lighting apparatus **105** based on input received from a user via controller **205** and/or based on predetermined algorithms or instructions. Controller **205** may include one or more printed circuit boards (PCBs) that may be used to control an operation of lighting apparatus **105**. One or more controllers **205** may control an operation of multiple lighting apparatuses **105** (e.g., a given controller **205** of a given lighting

apparatus **105** may control or be controlled by other lighting apparatuses **105** of lighting system **100**).

(20) Lighting assembly **200** may include one or more (e.g., a plurality of) lighting elements **210**. For example, lighting assembly **200** may include many (e.g., a multitude of) lighting elements **210**. Lighting element **210** may be any suitable type of lighting element such as, for example, any desired type of light-emitting diode (LED) light. Lighting element **210** may be an LED chip. Lighting element **210** may be any suitable type of lighting element such as, for example, an LED, an OLED, an incandescent lighting element, a fluorescent lighting element, and/or any other suitable type of lighting element.

(21) Mounting member **215** may be any suitable structural member for supporting controller **205** and lighting elements **210**. Mounting member **215** may be formed from any suitable thermally conductive material. Mounting member **215** may be a structural plate. Mounting member **215** may be a metal member. Mounting member **215** may be a thermally conductive member formed from any suitable thermally conductive material for supporting controller **205** and lighting elements **210** such as, for example, copper, chrome, brass, nickel, and/or aluminum material. For example, mounting member **215** may transfer heat away from one or more lighting elements **210**. As illustrated in FIG. 4, mounting member **215** may include one or more mounting apertures **220** for receiving portions of attachment assembly **400** for example as further described below. Mounting member **215** and/or attachment assembly **400** may act as a thermal transfer device to conduct heat away from lighting elements **210** and controller **205**.

(22) Returning to FIG. 3, sound absorbing assembly **300** may include one or more sound absorbing members **305**. Sound absorbing member **305** may be formed from any suitable material for absorbing sound such as, for example, foam (e.g., polyurethane foam), fiberglass, fabric or textile material (e.g., cloth, cotton, or felt), silicone, vinyl (e.g., mass loaded vinyl), caulk, polyester, green glue, cork, epoxy, and/or any other suitable sound absorbing material. Sound absorbing member **305** may be in tile form or any other suitable form, shape, or configuration. Sound absorbing member **305** may have any suitable thickness such as, for example, up to about 8" (e.g., or more), between about ½" and about 6", or between about 1" and about 3" (e.g., about 2"). As illustrated in FIG. 4, sound absorbing member **305** may include one or more absorbing apertures **310** for receiving portions of attachment assembly **400** for example as further described below.

(23) As illustrated in FIGS. 3 and 4, attachment assembly **400** may include one or more members **405** and a fastener **410**. Member **405** may be hollow or solid (e.g., partially solid and/or substantially entirely solid). Components of attachment assembly **400** may be formed from any suitable structural material for fastening such as, for example, metal, structural plastic, composite material, and/or any other suitable structural material. For example, components of attachment assembly **400** may be formed from steel (e.g., stainless steel), aluminum, titanium, brass, and/or any other suitable structural material. Member **405** may be configured to receive fastener **410**. Member **405** may be a tube member. Member **405** may include an elongated aperture **415** (e.g., hole) that may extend along a length of one or more members **405**. A surface of solid or hollow member **405** at which aperture **415** may be formed may be threaded or unthreaded. For example, the one or more hollow members **405** may be threaded or unthreaded tubes. Fastener **410** may be any suitable fastener such as a substantially entirely threaded fastener, a partially threaded fastener, or an unthreaded fastener. For example, fastener **410** may be a screw or a bolt that is threaded or unthreaded (e.g., or threaded along a partial length). Fastener **410** may have threads **420** that may or may not be received by corresponding threads of one or more members **405**. For example, in at least some exemplary embodiments, fastener **410** may be inserted into member **405** with or without engagement of threading (e.g., with or without being threaded into member **405**, or partially threaded into member **405**). For example in some exemplary embodiments, member **405** may or may not have threads at a surface forming aperture **415** (e.g., or the surface may be partially threaded). Member **405** may also have member threading **408** disposed at an outer surface of

member **405**. Fastener **410** may have a tip portion **425** that may be fastened into (e.g., received or screwed into) support structure **140**. For example, threads **420** may be disposed at tip portion **425** to facilitate attachment (e.g., screwing) of fastener **410** to support structure **140**. Fastener **410** may be attached to support structure **140** for example after attachment assembly **400** is attached to lighting assembly **200** and sound absorbing assembly **300** as described below. In at least some exemplary embodiments, fastener **410** may be attached to support structure **140** before or during attachment of attachment assembly **400** to lighting assembly **200** and/or sound absorbing assembly **300**.

(24) Attachment assembly **400** may also include a first positioning assembly **430** (e.g., a lower positioning assembly) having a plurality of attaching nuts **435**, a positioning nut **440**, and a positioning washer **445**. Attaching nuts **435**, positioning nut **440**, and positioning washer **445** may be any suitable threaded or unthreaded mechanical fasteners formed for example from the exemplary materials for components of attachment assembly **400** disclosed above. Attaching nuts **435** may have threading that may be configured to thread with member threading **408** that may be disposed at the exterior surface of member **405** (e.g., or threads **420** of fastener **410**). Attaching nuts **435** may be disposed on both sides of mounting member **215**). For example, members **405** may be attached to mounting member **215** at various points. Attaching nuts **435** may be threaded to tighten attaching nuts **435** against both sides of mounting member **215** (e.g., attaching nuts **435** may be tightened against each other and mounting member **215**). Mounting member **215** may also be attached to member **405** via any other suitable technique such as, for example, soldering or welding, adhesive, and/or any other suitable fasteners or mechanical technique. Positioning nut **440** may also have threading that may be configured to thread with member threading **408** that may be disposed at the exterior surface of member **405** (e.g., or threads **420** of fastener **410**). Positioning nut **440** may be threaded onto member **405** (e.g., or fastener **410**) to a desired position such that a desired empty space **145** may be provided between lighting assembly **200** and sound absorbing assembly **300**. Positioning washer **445** may be received on member **405** (e.g., or fastener **410**) between positioning nut **440** and sound absorbing member **305** (e.g., with sound absorbing member **305** resting on positioning washer **445** and positioning nut **440**). In at least some exemplary embodiments, positioning nut **440** may be formed as a hollow collar that slips over the exemplary disclosed member. Its length may establish an Empty Space distance of empty space **145**. For example, empty space **145** may be provided at any suitable distance such as, for example, between about 1" and about 12" or more, between about 2" and about 10", between about 3" and about 8", or any other desired distance. In at least some exemplary embodiments, empty spaces **145** (e.g., and/or other exemplary disclosed empty spaces for example as described herein) of adjacent lighting apparatuses **105** may be aligned (e.g., horizontally aligned or vertically aligned) for example as illustrated in FIG. 1.

(25) Returning to FIGS. 3 and 4, member **405** may be configured to have a similar diameter as a diameter of absorbing aperture **310** of sound absorbing member **305**. For example, member **405** may be relatively tightly or snugly received in absorbing aperture **310**. Sound absorbing member **305** may be placed on member **405** and pressed down until it rests on positioning washer **445** and positioning nut **440** (e.g., during an installation of lighting apparatus **105**).

(26) Attachment assembly **400** may also include a second positioning assembly **450** (e.g., an upper positioning assembly) having a holding nut **455** and a holding washer **460**. Holding nut **455** and holding washer **460** may be any suitable threaded or unthreaded mechanical fasteners formed for example from the exemplary materials disclosed above. Holding washer **460** may be placed on (e.g., received by) member **405** (e.g., or fastener **410**) so that holding washer **460** rests on sound absorbing member **305**. Holding nut **455** may have threading that may be configured to thread with member threading **408** that may be disposed at the exterior surface of member **405** (e.g., or threads **420** of fastener **410**). Holding nut **455** may be threaded onto member **405** (e.g., or fastener **410**) and suitably tightened against holding washer **460** to press holding washer **460** against sound absorbing

member **305** to hold sound absorbing member **305** in place (e.g., firmly in place). Holding nut **455** may be threaded onto member **405** (e.g., or fastener **410**) to a desired position such that a desired second empty space **150** is provided between sound absorbing assembly **300** and support structure **140**. Second empty space **150** may be provided at any suitable distance such as, for example, between about ½" and about 12" or more, between about 2" and about 10", between about 3" and about 8", or any other desired distance.

(27) In at least some exemplary embodiments, attachment assembly **400** may include a single member **405** that extends from support structure **140** to lighting assembly **200**. In at least some exemplary embodiments, attachment assembly **400** may include a plurality of members **405**. For example, attachment assembly **400** may include a first member **405** that extends from lighting assembly **200** to sound absorbing assembly **300** (e.g., from upper attaching nut **435** to positioning nut **440**), a second member **405** that extends through a thickness of sound absorbing member **305**, and a third member **405** that extends from sound absorbing assembly **300** to support structure **140** (e.g., from holding nut **455** to support structure **140**).

(28) In at least some exemplary embodiments, lighting apparatus **105** may be installed by first attaching lighting assembly **200** to attachment assembly **400** as described above, then attaching sound absorbing assembly **300** to attachment assembly **400** as described above, and then receiving fastener **410** through member **405**. For example after attachment assembly **400** is attached to lighting assembly **200** and sound absorbing assembly **300**, fastener **410** may be attached to support structure **140** by inserting fastener **410** through member **405** (e.g., that may already be attached to lighting assembly **200** and sound absorbing assembly **300** via first positioning assembly **430** and second positioning assembly **450**, respectively) from below and then fastening (e.g., screwing and tightening) tip portion **425** of fastener **410** to support structure **140**. For example, fastener **410** may be inserted into aperture **415** of member **405** with or without being threaded (e.g., fastener **410** may be inserted or slipped directly into aperture **415** and/or threadably inserted into aperture **415** along some or substantially an entire length of aperture **415**). Based on the exemplary disclosed configuration of lighting apparatus **105** described above, tightening forces that tighten fastener **410** to support structure **140** when fastener **410** is received in one or more members **405** may be transferred via member **405** (e.g., only exerted along a longitude of member **405** and/or fastener **410**) and not by other components of lighting apparatus **105**. Accordingly, lighting apparatus **105** may be attached to support structure **140** without attachment forces affecting lighting assembly **200** and sound absorbing assembly **300**.

(29) Lighting apparatus **105** may support sound absorbing assembly **300** without using a surround or similar attachment structure. For example based on attachment assembly **400** providing empty space **145** and/or second empty space **150**, an amount of exposed surface area of one or more sound absorbing members **305** may be increased based on the exemplary disclosed support configuration of attachment assembly **400**. As the exposed surface area may be increased (e.g., based on empty space **145** and/or second empty space **150**), a noise reduction capacity of lighting apparatus **105** may be increased. For example, substantially an entire surface area of one or more sound absorbing members **305** (e.g., except for areas covered by positioning washers **445** and holding washers **460**) may be exposed and may contribute to noise reduction performance of lighting apparatus **105**.

(30) Lighting apparatus **105** may be configured, disposed, and/or oriented in any desired manner relative to backlighting space **110** and occupied space **115**. For example, lighting apparatus **105** may be disposed above translucent layer **120** and occupied space **115** as illustrated in FIGS. **1** and **3**. Also for example as illustrated in FIG. **5**, lighting apparatus **105** may be disposed to a side of translucent layer **120** and occupied space **115**. For example, lighting apparatus **105** may act as an illuminated wall panel. Further for example, lighting apparatus **105** may be disposed underneath translucent layer **120** and occupied space **115** (e.g., when translucent layer **120** may be attached to glass or other transparent structural material).

(31) FIG. **6** illustrates a 2-dimensional bottom view of lighting apparatus **105**, which depicts one or

more exemplary disclosed open spaces **225** in lighting assembly **200** for allowing sound to pass through lighting assembly **200** to sound absorbing assembly **300**. As illustrated in FIG. 6, controller **205** and mounting member **215** may be configured to form one or more (e.g., a plurality of) open spaces **225**. Controller **205** and mounting member **215** may be formed with apertures corresponding to open space **225**. For example, controller **205** and mounting member **215** may be formed with apertures that are aligned so that one or more open spaces **225** may be formed through apertures of both controller **205** and mounting member **215** when controller **205** is attached to mounting member **215**. As illustrated in FIG. 6, open spaces **225** may be formed around members **405** of attachment assemblies **400** so that members **405** may attach lighting assembly **200** to sound absorbing assembly **300** and support structure **140**.

(32) One or more open spaces **225** may allow for sound to pass through lighting assembly **200**. Sound may thereby pass through lighting assembly **200**, via one or more open spaces **225**, to sound absorbing assembly **300** to be absorbed by one or more sound absorbing members **305**. Some or substantially all sound entering backlighting space **100** from occupied space **115** may thereby pass through lighting assembly **200** and be absorbed by sound absorbing assembly **300** without being reflected off of support structure **140** and/or other components of apparatus **105** and sent back into occupied space **115** (e.g., as an echo from backlighting space **110**).

(33) One or more open spaces **225** may also serve as apertures for electrical connectors **125** to pass through for connecting controller **205** to power source **130** and driver **135**. Electrical connectors **125** may thereby be disposed away from (e.g., above) an optical plane formed between lighting elements **210** and translucent layer **120** (e.g., a surface of translucent layer **120** facing backlighting space **110** and illuminated by lighting elements **210**). Electrical connectors **125** may thereby be routed through one or more open spaces **225** to reduce or substantially avoid interference of electrical connectors **125** with the illuminated plane of translucent layer **120**.

(34) FIG. 7 illustrates another exemplary embodiment of the exemplary disclosed apparatus. Apparatus **105a** may be generally similar to apparatus **105** and may include lighting assembly **200** and sound absorbing assembly **300** for example as described above that may be attached via an attachment assembly **400a** that may be generally similar to attachment assembly **400**. Attachment assembly **400a** may include first positioning assembly **430** and second positioning assembly **450** for example as described above. Attachment assembly **400a** may also include a hollow or solid member **405a** that may be generally similar to hollow or solid member **405**, and a fastener **410a** that may be generally similar to fastener **410**. Member **405a** and/or fastener **410a** may be received by a suspended fastener **465a**. For example, suspended fasteners **465a** may include a threaded aperture configured to receive either threading (e.g., similar to member threading **408**) on an outside surface of member **405a** or threads **420a** of fastener **410a**, and/or may be attached to member **405a** and/or fastener **410a** via any suitable fastener or mechanical attachment technique, adhesive, and/or any other suitable attachment technique. Suspended fastener **465a** may be attached to a suspension member **470a** via an eyelet **468a** and/or any other suitable mechanical attachment technique for attaching suspended fastener **465a** to suspension member **470a**. For example, suspended fastener **465a** may be a suspension ring (e.g., a threaded cable suspension ring). Suspension member **470a** may be any suitable structural member for supporting apparatus **105a** such as, for example, a cable, a wire, a rope (e.g., wire rope or fabric rope), a rod (e.g., a threaded rod or a tension rod), and/or any other suitable structural member. In at least some exemplary embodiments, suspension member **470a** that may be a suspension cable may be passed through eyelet **468a** and fastened such that apparatus **105a** may be positioned at a desired height or spacing and at a desired orientation (e.g., made level) relative to translucent layer **120**. Suspension member **470a** may also be passed through member **405a** and knotted or provided with any other suitable terminating device so that it will not pass back through member **405a**. Then it may be affixed to the exemplary disclosed structure as described above. This may eliminate eyelet **468a** in at least some exemplary embodiments. One or more suspension members **470a** may be attached to a support

structure (e.g., that may be similar to support structure **140**) via any suitable technique such as, for example, fasteners (e.g., anchor screws or bolts), welding, adhesive, and/or any other suitable attachment technique.

(35) Apparatus **105a** may thereby be suspended above translucent layer **120** for example as illustrated in FIG. 7. Apparatus **105a** may also include a spacer member **475a** that may include an aperture on which it may be received by member **405a** or fastener **410a** prior to member **405a** and/or fastener **410a** being attached to suspended fastener **465a**. For example, a clearance space may be provided between second positioning assembly **450** and suspended fastener **465a** to provide a space for spacer member **475a** to be received between second positioning assembly **450** and suspended fastener **465a** (e.g., or at any other suitable location of apparatus **105a**). Spacer member **475a** may be formed from material similar to components of attachment assembly **400** for example as described above. Spacer member **475a** may be a bracket including multiple apertures that may be received by a first member **405a** or fastener **410a** as illustrated above, and similar components of an adjacent apparatus **105a** (e.g., as illustrated in FIG. 7A). Accordingly and for example as illustrated in FIG. 7A, spacer member **475a** may be attached between adjacent apparatuses **105a** and may thereby provide a desired spacing between adjacent suspended apparatuses **105a**.

(36) FIG. 8 illustrates another exemplary embodiment of the exemplary disclosed apparatus. Apparatus **105b** may be generally similar to apparatus **105** and may include lighting assembly **200** and sound absorbing assembly **300** for example as described above that may be attached via an attachment assembly **400b** that may be generally similar to attachment assembly **400**. Attachment assembly **400b** may include first positioning assembly **430** and second positioning assembly **450** for example as described above. Attachment assembly **400b** may also include a member **405b** that may be generally similar to member **405**, and a fastener **410b** that may be generally similar to fastener **410**. Attachment assembly **400b** may also include one or more support legs **480b**. Support leg **480b** may be any suitable structural member formed for example from material similar to components of attachment assembly **400** as described above. Support leg **480b** may be a bent structural plate. A first support end portion **485b** of support leg **480b** may be received by member **405b** (e.g., or fastener **410b**) and/or attached via first position assembly **430** to lighting assembly **200**. Support leg **480b** may thereby be attached (e.g., firmly attached) to lighting assembly **200** and/or attachment assembly **400b**. A second support end portion **490b** of support leg **480b** may be supported at (e.g., supported on) and/or attached to a support structure **140b** that may be generally similar to the exemplary disclosed support structure described above. Support structure **140b** may be below the exemplary disclosed lighting assembly, or disposed above the exemplary disclosed lighting assembly. Apparatus **105b** may thereby be supported from above, level to, or from below by support structure **140b**. Varying lengths and/or shapes of support legs **480b** may be provided so that apparatus **105b** may be configured to be supported by a given configuration of support structure **140b**. Lighting assembly **200** and sound absorbing assembly **300** may thereby be supported above support structure **140b** and translucent layer **120**.

(37) The exemplary disclosed apparatus may be configured (e.g., fabricated) in any suitable size and shape to fit the contours of various ceilings and/or architectural features. For example, the exemplary disclosed apparatus may be configured to fit ceilings and/or architectural features including angles and curves. The exemplary disclosed apparatus may also be configured in various shapes and sizes to be used in any suitable type of lighting fixtures. For example, the exemplary disclosed apparatus may be included in pendant-type lighting fixtures.

(38) The exemplary disclosed system, apparatus, and method may be used in any suitable application involving lighting. For example, the exemplary disclosed system, apparatus, and method may be used in any suitable application for backlight lighting fixtures. The exemplary disclosed system, apparatus, and method may be used in any suitable application for absorbing sound and/or thermal transfer in lighting fixtures.

(39) FIG. 9 illustrates an exemplary process of using the exemplary disclosed apparatus. Process

500 begins at step **505**. At step **510**, the exemplary disclosed apparatus may be installed. For example as described above regarding apparatus **105**, lighting apparatus **105** may be installed by first attaching lighting assembly **200** to attachment assembly **400** as described above, then attaching sound absorbing assembly **300** to attachment assembly **400** as described above, and then receiving fastener **410** through member **405**. Fastener **410** may then be attached to support structure **140** for example as described above. Lighting apparatus **105a** may be similarly installed, including supporting lighting apparatus **105a** via suspension members **470a** for example as described above. Lighting apparatus **105b** may be similarly installed, including supporting lighting apparatus **105b** on support structure **140b** via one or more support legs **480b** for example as described above. As many exemplary disclosed lighting apparatuses as desired may be installed. In at least some exemplary embodiments, a complete factory assembly of exemplary disclosed LEDs, mounting plate, standoffs, sound absorbing material, and other exemplary disclosed components may be performed and then shipped as full units. Field installation may start with attaching the complete unit by any suitable mounting method.

(40) At step **515**, for example after installation of lighting apparatuses at step **510**, electrical connectors **125** may be installed through open spaces **225** for example as described above. In at least some exemplary embodiments, electrical connectors **125** connected to one or more drivers **135** may be routed above mounting members **215** and to desired locations (e.g., through open spaces **225**) at controllers **205** (e.g., attached to terminals on controllers **205**). Electrical connectors **125** may be bundled and tied to members **405** (e.g., and/or members **405a** and **405b**) as desired along their path from one or more drivers **135** to one or more apparatuses. In at least some exemplary embodiments, multiple lighting apparatuses may be supported by a single driver **135**.

(41) At step **520**, for example after the exemplary disclosed lighting apparatuses have been installed and electrically connected (e.g., connected via electrical connectors **125** and tested), translucent layer **120** (e.g., a translucent and sound penetrable film) may be installed below (e.g., or any other suitable location relative to) the exemplary disclosed lighting apparatuses (e.g., below or to the side of an LED PCB plane of lighting apparatus **105**, or below lighting apparatuses **105a** or **105b**). For example, translucent layer **120** may be installed based on attachment to support structure **140**, support structure **140b**, and/or attachment to any other suitable structural feature via mechanical fasteners, adhesive, and/or any other suitable attachment technique.

(42) At step **525**, after installation steps **510** through **520** are complete, the exemplary disclosed lighting apparatuses may operate to illuminate translucent layer **120** and occupied space **115** for example as described herein. Lighting apparatuses **105**, **105a**, and/or **105b** may operate so that light emanating from lighting elements **210** may illuminate (e.g., strike) translucent layer **120** (e.g., to uniformly illuminate translucent layer **120**). Translucent layer **120** may thereby illuminate occupied space **115**. Also for example, step **530** described below may precede step **525**.

(43) At step **530**, which may occur before, after, or simultaneously with step **525**, sound from occupied space **115** entering backlighting space **110** may be absorbed for example as described above. Sound within occupied space **115** may interact with translucent layer **120**. Sound may pass through the exemplary disclosed micro-perforations of translucent layer **120** for example as described above and/or may be transmitted via vibrations of translucent layer **120**. The sound may then enter backlighting space **115** and traverse the space between translucent layer **120** and the exemplary disclosed lighting assembly. The sound may pass through one or more open spaces **225** of lighting assemblies **200** for example as described above. The sound may then contact one or more sound absorbing members **305**, which may absorb (e.g., capture) the sound. Some or substantially all of the sound may thereby be captured and substantially prevented from bouncing off of the exemplary support structure and/or components of the exemplary disclosed lighting apparatuses and returning to occupied space **115**.

(44) At step **535**, it may be determined whether or not an operation of lighting system **100** should be continued. If operation of lighting system **100** is to be continued, process **500** may return to step

525. Steps 525 through 535 may be repeated for as many iterations as desired. Step 530 may precede step 525 as described above. If it is determined at step 535 that operation of system 100 is not to be continued, then process 500 ends at step 540.

(45) In at least some exemplary embodiments, a preassembled lighting apparatus may be raised into place, and the exemplary disclosed fastener may be driven through one or more hollow members and into the exemplary disclosed support structure. A substantially entire ceiling area may be covered with the exemplary disclosed lighting apparatuses. Translucent layer 120 may then be used to cover the exemplary disclosed lighting apparatuses.

(46) In at least some exemplary embodiments, lighting system 100 may provide one or more lighting apparatuses to both illuminate an area and absorb sound waves that intersect it, thereby reducing unwanted noise in the proximity of the one or more lighting apparatuses. Lighting system 100 may provide a backlight that uniformly illuminates (e.g., from behind the translucent material) translucent material covering a ceiling or other surfaces. Lighting system 100 may include one or more lighting apparatuses that secure an acoustic absorbent material, the acoustic absorbent material being separated from lighting elements of the one or more lighting elements.

(47) Lighting system 100 may provide for isolation of lighting assembly 200 from sound absorbing assembly 300, which in addition to increasing sound absorbing performance may also provide increased thermal transmittance (e.g., increasing a dissipation of heat generated by lighting elements 210 of lighting assembly 200). This characteristic of lighting system 100 may allow for higher efficiency of lighting assembly 200 and an ability to operate lighting elements 210 at higher output levels, thereby providing increased light to illuminate occupied space 115 at a relatively lower cost and use of energy.

(48) In at least some exemplary embodiments, lighting system 100 may provide for lighting assembly 200 and sound absorbing assembly 300 to be relatively easily mounted by screwing the exemplary disclosed lighting apparatuses directly to a support structure (e.g., a surface), suspending it from suspension members, or supporting lighting apparatuses from below. Also, based on isolation of sound absorbing assembly 300 from lighting assembly 200 based on providing empty space 145 (e.g., and/or second empty space 150), lighting system 100 may allow for electrical connectors (e.g., electrical connectors 125) to be installed after the exemplary disclosed lighting apparatuses are mounted, without electrical connectors interfering with an optical cavity (e.g., between lighting elements 210 and translucent layer 120 for example as described above). Accordingly, installing and operating lighting system 100 may not involve passing wires through sound-absorbing material and/or surface mounting material, which may be difficult, expensive, and time-consuming. Accordingly, lighting system 100 may involve simplified installation and/or service.

(49) In at least some exemplary embodiments, lighting system 100 may provide for relatively easily linking multiple lighting assemblies together in a suspended mounting arrangement that may provide substantially uniform light distribution. Further, lighting system 100 may allow for drivers 135 to be mounted with lighting assemblies 200 or mounted remotely. Additionally, lighting system 100 may allow for lighting elements 210 to provide static white light, tunable white light, and/or RGB or RGBW mixable light. Also, lighting system 100 may allow for lighting apparatuses to be relatively easily customized in shape and/or height.

(50) In at least some exemplary embodiments, the exemplary disclosed lighting elements and acoustic panels of lighting system 100 may be sized according to any desired shape (e.g., cut to different shapes). Also for example, the exemplary disclosed lighting apparatuses may be spaced so that they are touching to provide a continuous layer of sound absorbing material and even illumination of the exemplary disclosed translucent material.

(51) In at least some exemplary embodiments, the exemplary disclosed method may include providing a lighting assembly (e.g., lighting assembly 200) including at least one lighting element, attaching at least one attachment assembly (e.g., attachment assembly 400, attachment assembly

400a, or attachment assembly **400b**) to the lighting assembly, forming an empty space between the lighting assembly and a sound absorbing assembly (e.g., sound absorbing assembly **300**) based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member, and illuminating a translucent or transparent layer using the at least one lighting element. The exemplary disclosed method may also include receiving sound through the translucent or transparent layer and at least one open space formed in the lighting assembly, and absorbing the sound with the at least one sound absorbing member. Attaching the at least one attachment assembly to the lighting assembly and the sound absorbing assembly may include attaching at least one member of the at least one attachment assembly to the lighting assembly and the sound absorbing assembly. The exemplary disclosed method may further include attaching the at least one attachment assembly to a support structure based on inserting a fastener through the at least one member and attaching the fastener to the support structure. Attaching the at least one attachment assembly to the support structure may include transferring tightening forces from the fastener to the at least one member without transferring the tightening forces to the lighting assembly or the sound absorbing assembly. Forming the empty space between the lighting assembly and the sound absorbing assembly may include tightening attaching nuts received on the at least one member against the lighting assembly, and threading a positioning nut on the at least one member against a positioning washer on which the at least one sound absorbing member rests. The exemplary disclosed method may also include providing a second empty space between the sound absorbing assembly and the support structure, wherein the sound absorbing assembly may be disposed between the support structure and the lighting assembly, and separates the empty space from the second empty space. Attaching the at least one attachment assembly to the sound absorbing assembly may include inserting at least one member of the at least one attachment assembly through an aperture of the at least one sound absorbing member. The exemplary disclosed method may further include suspending the at least one attachment assembly above the translucent or transparent layer from at least one suspension member. The exemplary disclosed method may also include supporting the lighting assembly with a support structure disposed above the translucent or transparent layer. The exemplary disclosed method may further include electrically connecting the lighting assembly with a driver via an electrical connector that passes through an open space formed by a printed circuit board and a mounting member of the lighting assembly that may be attached together. The exemplary disclosed method may also include transferring heat away from the at least one lighting element that may be at least one LED, wherein the at least one LED provides at least one of static white light, tunable white light, or RGB or RGBW mixable light.

(52) In at least some exemplary embodiments, the exemplary disclosed lighting apparatus may be a lighting apparatus for illuminating a translucent or transparent layer. The exemplary disclosed lighting apparatus may include a lighting assembly (e.g., lighting assembly **200**) including at least one lighting element configured to illuminate the translucent or transparent layer, a sound absorbing assembly (e.g., sound absorbing assembly **300**) including at least one sound absorbing member, and at least one attachment assembly (e.g., attachment assembly **400**, attachment assembly **400a**, or attachment assembly **400b**) attached to the lighting assembly and the sound absorbing assembly so that an empty space is formed between the lighting assembly and the sound absorbing assembly. The at least one attachment assembly may include at least one member, the at least one member configured to be fastened to the lighting assembly and to the sound absorbing assembly. The at least one attachment assembly may include a member and a fastener, the fastener configured to be received by the member and fastened to a support structure. The lighting assembly may include a mounting member and a controller attached to the mounting member, the mounting member and the controller forming at least one open space when attached.

(53) In at least some exemplary embodiments, the exemplary disclosed method may include providing a plurality of lighting apparatuses, each of the plurality of lighting apparatuses including

at least one attachment assembly (e.g., attachment assembly **400**, attachment assembly **400a**, or attachment assembly **400b**) attached to a lighting assembly (e.g., lighting assembly **200**) and a sound absorbing assembly (e.g., sound absorbing assembly **300**) so that an empty space may be formed between the lighting assembly and the sound absorbing assembly of each of the plurality of lighting apparatuses, illuminating a translucent or transparent layer using the lighting assemblies of the plurality of lighting apparatuses, the lighting assemblies each including at least one open space, receiving sound through the translucent or transparent layer and the at least one open spaces formed in the lighting assemblies of the plurality of lighting apparatuses, and absorbing the sound with the sound absorbing assemblies of the plurality of lighting apparatuses. The exemplary disclosed method may also include spacing at least some of the plurality of lighting apparatuses from each other based on attaching a spacing member between the at least one attachment assemblies of adjacent lighting apparatuses of the plurality of lighting apparatuses. The exemplary disclosed method may further include electrically connecting the lighting assemblies of the plurality of lighting apparatuses with a driver via at least one electrical connector that passes through the at least one open spaces formed in the lighting assemblies of the plurality of lighting apparatuses. The translucent or transparent layer may be a ceiling film and the empty spaces of the plurality of lighting apparatuses are horizontally aligned.

(54) In at least some exemplary embodiments, the exemplary disclosed system, apparatus, and method may provide an efficient and effective system for absorbing sound in lighting apparatuses (e.g., lighting fixtures). For example, the exemplary disclosed system, apparatus, and method may provide for efficient dissipation of heat generated by a lighting fixture, which may allow for higher efficiency of the lighting fixture and the ability to provide increased illumination at a relatively lower cost. The exemplary disclosed system, apparatus, and method may also provide a lighting fixture that may be quickly and easily installed.

(55) It will be apparent to those skilled in the art that various modifications and variations can be made to the exemplary disclosed system, apparatus, and method. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the exemplary disclosed apparatus, system, and method. It is intended that the specification and examples be considered as exemplary, with a true scope being indicated by the following claims.

Claims

1. A method, comprising: providing a lighting assembly including at least one lighting element; attaching at least one attachment assembly to the lighting assembly; forming an empty space between the lighting assembly and a sound absorbing assembly based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member; and illuminating a translucent or transparent layer using the at least one lighting element; wherein attaching the at least one attachment assembly to the lighting assembly and the sound absorbing assembly includes attaching at least one member of the at least one attachment assembly to the lighting assembly and the sound absorbing assembly; and wherein the at least one member of the at least one attachment assembly is hollow.
2. The method of claim 1, further comprising receiving sound through the translucent or transparent layer and at least one open space formed in the lighting assembly, and absorbing the sound with the at least one sound absorbing member.
3. The method of claim 1, further comprising attaching the at least one attachment assembly to a support structure based on inserting a fastener through the at least one member and attaching the fastener to the support structure.
4. The method of claim 3, wherein attaching the at least one attachment assembly to the support structure includes transferring tightening forces from the fastener to the at least one member without transferring the tightening forces to the lighting assembly or the sound absorbing assembly.

5. The method of claim 3, further comprising providing a second empty space between the sound absorbing assembly and the support structure, wherein the sound absorbing assembly is disposed between the support structure and the lighting assembly, and separates the empty space from the second empty space.
6. The method of claim 1, wherein forming the empty space between the lighting assembly and the sound absorbing assembly includes tightening attaching nuts received on the at least one member against the lighting assembly, and threading a positioning nut on the at least one member against a positioning washer on which the at least one sound absorbing member rests.
7. The method of claim 1, wherein attaching the at least one attachment assembly to the sound absorbing assembly includes inserting at least one member of the at least one attachment assembly through an aperture of the at least one sound absorbing member.
8. The method of claim 1, further comprising suspending the at least one attachment assembly above the translucent or transparent layer from at least one suspension member.
9. The method of claim 1, further comprising supporting the lighting assembly with a support structure disposed above the translucent or transparent layer.
10. The method of claim 1, further comprising electrically connecting the lighting assembly with a driver via an electrical connector that passes through an open space formed by a printed circuit board and a mounting member of the lighting assembly that are attached together.
11. The method of claim 1, further comprising transferring heat away from the at least one lighting element that is at least one LED, wherein the at least one LED provides at least one of static white light, tunable white light, or RGB or RGBW mixable light.
12. A lighting apparatus for illuminating a translucent or transparent layer, comprising: a lighting assembly including at least one lighting element configured to illuminate the translucent or transparent layer; a sound absorbing assembly including at least one sound absorbing member; and at least one attachment assembly attached to the lighting assembly and the sound absorbing assembly so that an empty space is formed between the lighting assembly and the sound absorbing assembly; and wherein the at least one attachment assembly includes at least one member configured to attach the at least one attachment assembly to the lighting assembly and the sound absorbing assembly; and wherein the at least one member is hollow and configured to receive a fastener.
13. The lighting apparatus of claim 12, wherein the lighting assembly is a surface mount lighting fixture or a pendant lighting fixture.
14. A method, comprising: providing a lighting assembly including at least one lighting element; attaching at least one attachment assembly to the lighting assembly; forming an empty space between the lighting assembly and a sound absorbing assembly based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member; illuminating a translucent or transparent layer using the at least one lighting element; wherein attaching the at least one attachment assembly to the lighting assembly and the sound absorbing assembly includes attaching at least one member of the at least one attachment assembly to the lighting assembly and the sound absorbing assembly; and attaching the at least one attachment assembly to a support structure based on inserting a fastener through the at least one member and attaching the fastener to the support structure.
15. The method of claim 14, further comprising receiving sound through the translucent or transparent layer and at least one open space formed in the lighting assembly, and absorbing the sound with the at least one sound absorbing member.
16. The method of claim 14, wherein attaching the at least one attachment assembly to the support structure includes transferring tightening forces from the fastener to the at least one member without transferring the tightening forces to the lighting assembly or the sound absorbing assembly.
17. The method of claim 14, wherein forming the empty space between the lighting assembly and the sound absorbing assembly includes tightening attaching nuts received on the at least one

member against the lighting assembly, and threading a positioning nut on the at least one member against a positioning washer on which the at least one sound absorbing member rests.

18. The method of claim 14, further comprising providing a second empty space between the sound absorbing assembly and the support structure, wherein the sound absorbing assembly is disposed between the support structure and the lighting assembly, and separates the empty space from the second empty space.

19. The method of claim 14, wherein attaching the at least one attachment assembly to the sound absorbing assembly includes inserting at least one member of the at least one attachment assembly through an aperture of the at least one sound absorbing member.

20. A method, comprising: providing a lighting assembly including at least one lighting element; attaching at least one attachment assembly to the lighting assembly; forming an empty space between the lighting assembly and a sound absorbing assembly based on attaching the sound absorbing assembly to the at least one attachment assembly, the sound absorbing assembly including at least one sound absorbing member; and illuminating a translucent or transparent layer using the at least one lighting element; wherein attaching the at least one attachment assembly to the lighting assembly and the sound absorbing assembly includes attaching at least one member of the at least one attachment assembly to the lighting assembly and the sound absorbing assembly; and wherein forming the empty space between the lighting assembly and the sound absorbing assembly includes tightening attaching nuts received on the at least one member against the lighting assembly, and threading a positioning nut on the at least one member against a positioning washer on which the at least one sound absorbing member rests.
